

Cover letter — *Quarterly Journal of Economics*

Editor
Quarterly Journal of Economics

Dear Editor,

I am submitting “The Long Shadow of Seasonality: Climate Endowments, Agricultural Pathways, and Escape from the Malthusian Trap” for consideration at the *QJE*.

The paper extends *matranga2024* into the post-Neolithic period and produces three findings that I believe make it an appropriate companion to that work. The argument is that the climate dimension *Matranga* identified at the origin of agriculture also shapes which agricultural system a society adopts after the Neolithic, and that the selected system in turn governs the Malthusian feedback and the response to climate shocks all the way to the modern transition.

Empirical contributions. *First*, I construct a country-level and sub-national monthly climate panel reaching back to 1421 CE by area-weighting the ModE-RA paleo-reanalysis onto HYDE 3.5 sub-units. The country-month correlation against country-level ERA5 (which I rebuild from raw 0.25° fields onto HYDE boundaries) is 0.971 in median; the annual anomaly correlation after detrending is 0.938. This is, to my knowledge, the first dataset of its kind for econometric work and substantially expands what can be tested in the climate-and-pre-industrial-growth literature.

Second, I show that raw temperature seasonality alone, the natural *Matranga* measure, does *not* significantly distinguish agricultural pathways at the country level once climate is measured properly ($F = 0.50$, $p = 0.68$ on $N = 153$ countries). What does is the productive-months index—the joint structure of temperature and precipitation that determines whether rain-fed agriculture is feasible ($F = 6.31$, $p < 0.001$). This is a sharpening of the *Matranga* story, not a refutation: storage demand is generated by the joint constraint of growing-season conditions and the off-season’s harshness, not by temperature variability in isolation. I view the *QJE*-natural framing as “*Matranga*’s storage-demand mechanism, sharpened and extended into the post-Neolithic.”

Third, exploiting HYDE’s sub-national grid (3,187 sub-units across 196 countries) and adding country fixed effects, I identify the storage-demand effect on agricultural specialisation off purely within-country variation. The within-country point estimate is $3.5\times$ the between-country estimate, consistent with cross-country regressions being attenuated by mean-reverting institutional controls. Within-country, an extra productive month per year is associated with 6.7 pp higher crop share at 1750 ($p = 0.019$ with country-clustered SE, $p = 0.0001$ with two-way clustering by country and 10° latitude band) and 0.66 log-units higher population density ($p < 10^{-3}$). This is the paper’s tightest causal evidence and the move that puts the long-run climate-development literature on a within-country footing.

A 1421–1950 panel with real annual climate also surfaces two by-product findings: a within-paper signature of the demographic transition (the Malthusian density coefficient progressively vanishing across subperiods) and a sign-flip on inter-annual temperature volatility between 1750–1900 and 1900–1950 (-0.0125 , $p < 10^{-4}$), consistent with modern monoculture being more volatility-sensitive than diversified pre-industrial systems.

The paper is single-authored and the empirical infrastructure (ModE-RA aggregation, country-level ERA5 rebuild, sub-national panels) is built from scratch. A replication package will be deposited on OSF upon acceptance. The work has not been previously published and is not under consideration elsewhere; I declare no conflicts.

Sincerely,

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Suggested referees (independent of the author): A. Matranga (NES); O. Galor or D. Weil (Brown); Q. Ashraf (Williams); N. Voigtländer (UCLA); S. Hsiang (Stanford). I will note that the empirical strategy speaks most directly to readers familiar with Matranga (2024) and Ashraf–Galor (2011).